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From human to urban metabolism: mapping flows of excreted nutrients based on current and prospective diets

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Abstract

Increasing awareness of the health, environmental, and geopolitical issues associated to mineral fertilizers have brought renewed attention to alternative sources of nutrients, such as those that flow from the sanitation systems of cities. This raises the questions: how much can we recover from source-separated human urine and feces? Where are the largest deposits? Where should we start? To answer these as precisely as possible — taking into account the specificity of the area of interest, such as local diets or population pyramid — this work proposes a detailed model of the local territorial metabolism and the associated flows of nitrogen and phosphorus. As it can be adapted for any country to assess deposits at various scales, from national down to the individual building, it can support both policymakers and local urban planners.

The model consists in the combination of equations describing human metabolism, information on the local diet, and open statistics on the populations inhabiting the area. Its results are validated against sanitation data from two areas in France and the Netherlands and compared to previous works in the scientific literature. We also demonstrate how the model can be used to inform prospective analyses on the evolution of the load entering sanitation systems and the nitrogen-to-phosphorus ratios, depending on population and dietary changes. Finally, we discuss how this open-source model can assist local actors in planning projects to separate human excreta and produce local and recycled fertilizers to increase their food security and autonomy.

Keywords: nitrogen, phosphorus, fertilizer, sanitation, dietary shifts, source separation, circular bioeconomy, urban planning

List of abbreviations

N: Nitrogen

P: Phosphorus

WWTP: WasteWater Treatment Plant

PE: Population Equivalent (60 gBOD5/day)

BOD5: 5-day Biological Oxygen Demand

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1. Introduction

Human excreta have long been used as fertilizers for agriculture, notably before the widespread use of synthetic fertilisers in the 20th century with the industrial and agricultural revolutions, with 40% nitrogen recycling reached in Paris in the 19th century, and up to 70% in some Swedish areas (Esculier 2018). The development of centralized sanitation systems during this century, associated with the drastic reduction of synthetic fertilizers' cost in western countries lead to a transition to a more linear management of nutrient flows (Le Noë et al. 2018; Barles 2010). This new linear management contributes significantly to the perturbation of nitrogen (N) and phosphorus (P) biogeochemical cycles (Richardson et al. 2023). With the increasing concerns of environmental issues and securing fertilizer supply chains, works in urban metabolism (Esculier et al. 2019) and circular economy (Moya et al. 2019; Lehtoranta et al. 2022a) put the recycling of human excreta back on the scientific and technical agenda. Some studies even question the relevance of modern sanitation systems, calling for new socio-technical organizations centred on ecological sanitation (Langergraber and Muellegger 2005; Larsen et al. 2009; Larsen et al. 2021). Following the first studies, several analyses illustrated the potential benefits of recycling urine at a territorial scale, particularly where the resource is important, e.g. around dense urbanized areas or where the agricultural demand for fertilizer is important (Lehtoranta et al. 2022a; Wassenaar et al. 2023; Kleckers et al. 2024). Other studies aimed at identifying the spatial distribution of excreta production to imagine strategies of urban development dedicated to collecting the associated nutrient flows. For instance, Wielemaker et al. (2020) identified hotspots of urine production at the building scale in Amsterdam (Netherlands), Gunnarsson et al. (2023) assessed urine separation in a new development area in Malmö (Sweden) and Crane et al. (2024) investigated urine source separation at the community-level for commercial and institutional buildings in Phoenix (USA).

Such studies must evaluate the total amount of excreta available in a given territory and the nutrients that could be contained within this resource, which has led to the use of a large variety of models. Gunnarsson et al. (2023) and Kleckers et al. (2024) used the volume production and average excretion of nitrogen and phosphorus, based on wastewater data. Other studies used human metabolic data, either computing excretion values based on the medical literature for adults only (Wielemaker et al. 2020) or using FAO data to estimate the average nutrient ingestion from food for the whole population, assuming that the total amount of nutrients ingested was excreted through urine and feces (Mihelcic et al. 2011; Perez-Mercado et al. 2022; Wassenaar et al. 2023; Kleckers et al. 2024). However, given that nutrients are lost through other flows than urine and feces, these studies tend to overestimate the actual deposits and do not necessarily distinguish the amount of nutrients excreted in urine from those excreted in feces. Some studies tried to further refine the data used for their metabolic models: Lehtoranta et al. (2022a) considered the fraction of children in the local population, to adapt the nutrient excretion accordingly, while Crane et al. (2024) used data on urine volumes, coupled to data on the variation of phosphorus concentration depending on the gender and the time of excretion during the day (higher concentration in the first morning urine compared to the rest of the day).

The methods and models used in these studies usually provide reasonable estimates of the resources available at the territorial scale, which are relevant to inform large-scale public policies. However, most fall short when it comes to informing local decision makers and developing detailed territorial strategies that include alternative sanitation systems. Indeed, urban planning projects must dimension the logistics chains and transformation systems to assess the expected costs and benefits in terms of avoided impacts for the wastewater treatment plants (WWTPs) and fertiliser production. To do so, they need the ability to a) estimate as precisely as possible how much nutrients are contained in the urine and in feces separately, since only one type of excretion might be collected in the project, and b) localize excreta generation, since the type of building will influence the deposits (excretions in an office will differ from those of a residential building).

Regarding the relevance of the excretion values, previous studies have typically used values from a limited amount of sources[1] for daily volume or nutrient production, with little to no discussion of

1. notably Rose et al. (2015), Vinnerås et al. (2006) and Jönsson et al. (2004) (archived PDF)

their relevance to the specific case study. Unfortunately, most of these figures are specific to a given population, in a given area, and not linked to human metabolism data. Since excretion values can vary significantly, even within the normal range for humans (as shown in appendix Figure A3), choosing physiologically plausible but unconstrained values for urine and feces, can lead to studies using a total phosphorus excretion which is twice the average amount consumed through food and beverages in the area (Zhou et al. 2026). The remaining studies, that do constrain excretions by the nutrient intakes from the local diet, either do not differentiate each flow or were not designed to account for some important features of the local populations (age, activities, holidays...).

Let us given a concrete example: what is the maximum potential of urine source-separation if it were implemented in all schools of a given area? To answer this question, the quantification model should account for the specific place of production of the excreta, the specific populations involved, and distinguish between urine and feces. Though some studies (Besson et al. 2021; Lehtoranta et al. 2022b; Crane et al. 2024) took into account the location, they used simplified ratios based on the time spent at home or at work, while excretions tend to happen at specific times (e.g. when getting up or before going to bed) and not uniformly throughout the day (Burgio et al. 1991; Heaton et al. 1992). Apart from Lehtoranta et al. (2022a), most studies did not use values that account for local specificities in terms of population structure (children do not excrete the same amount as adults) and can therefore not reliably estimate the deposits in schools. Overall, none of the current methods are well suited to answer this question — especially not in a way that can be adapted to different countries of the world — because they do not combine the necessary dietary, age, and presence data.

There is therefore a need for models that can, in a straightforward manner, enable to determine which values should be used to assess nutrient excretion in urine and feces in an area given local specificities, such as dietary intake or population pyramid. These models should support the analysis of urban or territorial metabolism at various scales (from bioregions to individual buildings) in order to support decision-making processes for urban planning or local development projects. Though source-separation projects will need to go further and assess how much can actually be collected, then recovered in practice (depending on the specific toilets and processes used), an initial assessment of the generated nutrients (what we call "deposits") will always be necessary. In this article, we therefore propose a model, based on medical data on human metabolism, which a) uses metabolic constraints to ensure that there is no excessive variability in the results, b) distinguishes between urine and feces, c) accounts for population flows and time patterns to assess both where each type of excretion occurs during the day and how the presence or absence of certain populations (e.g. during holidays) impacts the deposits, c) accounts for age, gender, and dietary changes to provide more precise data about the nutrient deposits. Excreted volumes of urine are also modelled from medical data and can be used to estimate nutrients' concentrations in a second step.

Furthermore, by connecting the metabolic model directly to local diets, one can not only improve the precision of the results, but also analyse how the stakes of source separation and those associated to dietary changes could be connected. Indeed, source separation connects several sectors such as sanitation, agriculture, water and energy production, and can be used as a way to simultaneously reduce the environmental impacts associated to all four sectors. To meet the challenges of the climate, environmental and public health crises, studies have shown the importance of changing agricultural but also dietary habits, which are a major driver of environmental and health impacts (Campbell et al. 2017; Aydodu and Gezmen Karada 2025; Rockström et al. 2025; Perraud et al. 2025). Among the necessary dietary changes, reducing the overconsumption in the global North (especially from animal-based products) and incorporating a larger fraction of plant-based products are crucial steps to reduce environmental impacts and enable countries or regions to guarantee their food security and autonomy (Billen et al. 2021). As these dietary changes affect the amount of nutrients ingested and excreted, a model integrating prospective diets can assess the impact of those dietary shifts on sanitation and nutrient recovery from excreta.

In this study, we therefore provide a new method to quantify the amount of nutrients in excreta, based on the "balance" principle and detailed medical and dietary data. We also use medical and socio-

technical data to model when and where excretions occur. These two methods are then integrated into a single model which accounts for both human metabolism and the population flows underlying urban or territorial metabolism. This enables to consistently quantify the amount of nutrients in urine and feces and map how they are distributed spatially, down to the building's resolution. We illustrate the model's results on two case studies: the city of Toulouse and its surroundings municipalities, in France, and Amsterdam, in the Netherlands. In both cases, we discuss the potential impacts of source-separation and dietary changes on the local sanitation and agricultural systems, and how these could contribute to lowering the impact of future population increase as well as improving local food autonomy.

2. Methods

All codes used in the article are detailed and available in the following repository, where the main notebook can be viewed online[2]: https://forge.inrae.fr/tfardet_code_articles/human_urban_metabolism_2026

2.1. Modelling human NP metabolism

To assess the production of nitrogen (N) and phosphorus (P) excreted by a population, a metabolic model is proposed to quantify the amounts excreted in urine, feces, and other body losses or retentions, as a function of the dietary intake. Since the concentration of nutrients in urine can also be of interest during the investigation of practical cases of urban metabolism, a model of urinary excretion is also provided in the Appendix.

The foundation of the model is the mass-conservation principle: over a century of medical research has confirmed that there is an almost exact "balance" between the nutrients entering and exiting the human body, except when there is a quick mass gain or loss, e.g. during growth spurts for children and adolescents (Rand et al. 2003). Compared to previous "stoechiometric" models (Muñoz et al. 2007; Vermeulen et al. 2023), we use medical data, to allocate the fraction of both nitrogen and phosphorus in all bodily flows: urine, feces, other losses, and variation of the amount stored in the body. For adults, the model uses data from balanced studies that simultaneously quantify, over a period of time, nutrient ingestion and excretion (in urine and feces separately, as well as through other bodily flows in some studies). For children and adolescents, we estimate the amount stored in the body due to growth through official statistics on mass-gain curves and average body compositions in nitrogen and phosphorus.

The sources used to estimate all parameters and functions is given in Appendix A. The model itself and the data that has been used to validate it are provided through the Python library OrgMatt (Fardet 2026). The data is freely available through CSV files at the following address: https://git.sr.ht/~tfardet/OrgMatt/tree/main/item/orgmatt/data.

The metabolic model accounts for the flows entering and leaving the humans body. The amount of a nutrient contained in each flow is defined as:

- m_i, for the mass of nutrient ingesting from the dietary intake,
- m_u, for the mass excreted in urine,
- m_f, for the fecal excretion,
- m_l, for the nutrient mass lost from sweat, breathing, hair or nails, skin, and sexual secretions,
- m_r, for the nutrients retained in the body, notably for children during growth.

2. see https://codeberg.org/tfardet/article_human_urban_metabolism if a backup is needed.

Since the total mass of each nutrient $n \in \{\text{N, P}\}$ is conserved, the previous flows can be connected by the following equation:

$$m_i^{(n)} = m_u^{(n)} + m_f^{(n)} + m_l^{(n)} + m_r^{(n)} \tag{1}$$

As the equations are the same for both nutrients, we will hide the n to make them more readable and thus use $m_i = m_u + m_f + m_l + m_r$ as an equivalent for Equation 1 and only add the nutrient when providing specific values for N or P.

For agricultural purposes, the larger and more easily recoverable part of the nutrients is contained in urine and feces. The main objective of the metabolic model is thus to estimate the amount of nutrient present in these two flows as a function of the dietary intake:

$$m_u = F_u(m_i) \quad \text{and} \quad m_f = F_f(m_i) \tag{2}$$

The shape of these functions, as well as the other losses, were obtained from results from the medical literature, using studies that simultaneously quantified the intake, urinary and fecal excretions of each nutrient for different age groups. The code and data (including the list of studies used) are provided in the Appendix, section D. Based on 10 studies examining a range of healthy intake values, a linear relationship could be established between intake, urinary, fecal excretions, and losses (see Figure 1 A and B). In the rest of the study, all relations will therefore be considered linear with respect to the nutrient intake and modelled as follow each flow $x \in \{\text{u, f, l, r}\}$:

$$m_x = a_x m_i. \tag{3}$$

Few results were available to assess nutrient retention in children in the data mentioned previously. The retention for each nutrient was therefore estimated (based on 12 medical studies) using the mass fraction of nutrient in the body, f_{bm}, multiplied by the average change in mass at a given age (Δm_{body}) (based on national growth curves). The mass of nutrient n retained in the body is thus computed as:

$$m_r = f_{bm} \cdot \Delta m_{\text{body}} = a_r m_i \quad \Rightarrow \quad a_r = \frac{f_{bm} \cdot \Delta m_{\text{body}}}{m_i} \tag{4}$$

For children, the model must take into account the nutrients retained in growing bones and tissues. To be retained in the body, nutrients must first be metabolised (that is, absorbed in the digestive tract and become available to the body). This means that nutrients that are retained in the body or excreted in urine stem from the same pool of absorbed nutrients while those excreted in feces come from the non-metabolised part combined with a stable amount of endogenous nitrogen from physiological digestive secretions (Fuller and Reeds 1998). Nutrient absorption, estimated through balance studies, shows high variability even among adults of the same age (Rand et al. 2003) and, while a decreasing trend is reported for elderly people, we could not find reports of a variation between adults and children. An analysis of the nitrogen content of feces in children of different ages, thus associated to different growth rates, did not display significant differences, which is also consistent with the hypothesis of a constant absorption rate (Rivero-Marcotegui et al. 1998). Because of this, the hypothesis used in the model, is that nutrient absorption in the body is identical between children and adults, and thus that there is a reduced nutrient content in the urine of children because part of the absorbed nutrients are retained in bones or tissues.

For physiological diets, the final model is thus constructed as follow, using linear relations:

1. The amount of nutrient excreted in feces is computed as $m_f = a_f m_i$, using the value given in Figure 1A and B, and is assumed to be age-independent.

2. As adults display negligible retention when their weight is stable, the fraction of nutrient in various losses is estimated as $m_l = a_l m_i = (1 - a_f - a_{u,\text{adult}})\, m_i$, using the values detailed in the Results section, and is assumed to be age-independent.

3. For children and teenagers, the fraction of retention in the body is estimated for each age j as $m_{r,j} = a_{r,j} m_{i,j}$, as shown on Figure 1C; a_r is set to zero for adults and seniors.

4. Finally, the age-dependent excretion in urine is deduced via $m_{u,j} = a_{u,j} m_{i,j} = (1 - a_f - a_{r,j} - a_l) m_{i,j}$. For children and teenagers, this results in a fraction that is lower than $a_{u,\text{adult}}$.

2.2. Dietary intakes

Since the excretion of N and P are directly related to the dietary intakes of the population, diets must be assessed to quantify the nutrients that could be collected in a given area. Furthermore, as future socio-economic pathways may promote nutrient recycling and more plant-based diets, we also include prospective diets to assess how nutrient deposits in excreta may evolve in such pathways.

In this study, the following dietary intakes were used to estimate nutrient deposits:

- The most recent available diet for France, from the INCA3 study (ANSES 2017). This study assessed the food consumption habits of the French population in 2014 and 2015 and translated the food consumption data into nutrients intake for multiple age groups. It is used to estimate current excretions in the Toulousian area of France, to validate the model.

- Dutch diets from the Food Consumption Survey (VCP) 2012–2016 and 2019–2021 of the National Institute for Public Health and the Environment (RIVM), by (Van Rossum et al. 2020; Van

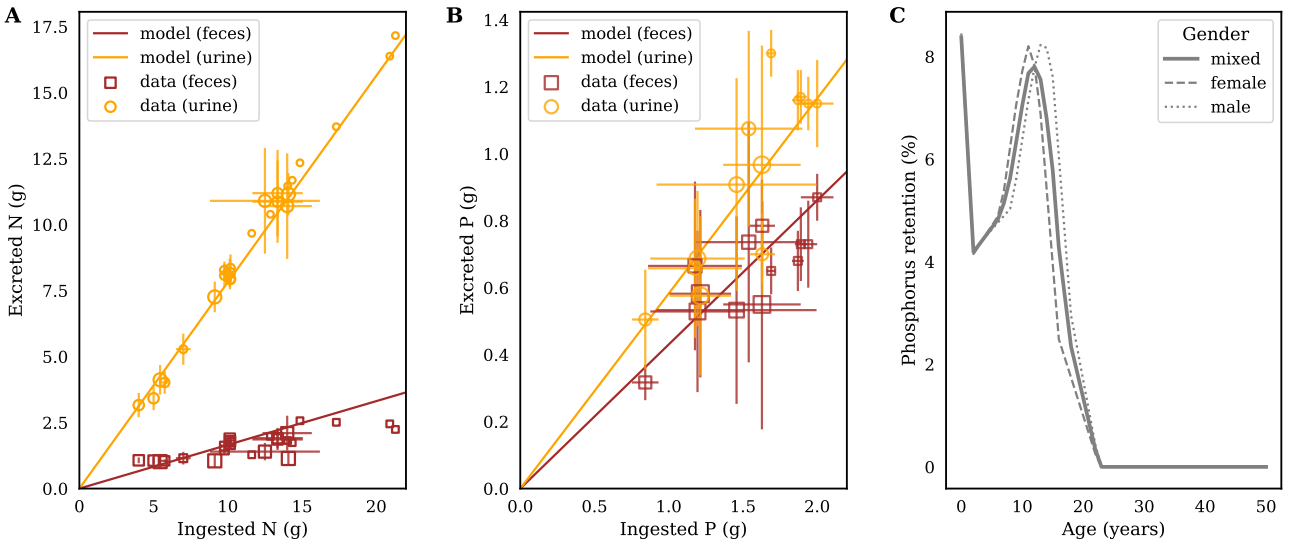


Figure 1: Relationship between ingested, excreted and retained nitrogen (N) and phosphorus (P) for adults and children. The yellow and brown lines represent the nutrients excreted in urine and feces respectively. Data points for urine and feces are shown in yellow and brown; their size is correlated to the number of patients in the study, the error bars show the range of values. **A.** For nitrogen, in physiological diets ($n_i > 0$), the relationship can be approximated by a simple linear relation. Using a weighted orthogonal distance regression (ODR) to account for the uncertainty in the data, we obtained the best fits with a slope of $a_u^{(N)} = 0.782 \pm 0.003$ for urine ($R^2 = 1$) and $a_f^{(N)} = 0.166 \pm 0.006$ for feces ($R^2 = 0.97$). **B.** In the case of phosphorus, the best fit of the weighted ODR is $a_u^{(P)} = 0.58 \pm 0.03$ for urine ($R^2 = 0.93$) and $a_f^{(P)} = 0.43 \pm 0.02$ for feces ($R^2 = 0.79$). **C.** For children and young adults, the excretion in urine will be reduced as part of the digested nutrients remain stored in the body (retention) due to growth and bone consolidation, as shown here for phosphorus. Average retention model is shown as a grey full line, retention for female and male individuals is shown by the dashed and dotted lines respectively. It shows two peaks, consistent with the two growth spurts periods for infants and adolescents (Hauspie and Roelants 2012).

Rossum et al. 2023); they are used to estimate the excretions in the city of Amsterdam and validate the model.

- Prospective French diets from the Afterres2050 scenario, which proposes a trajectory for the agricultural and food transition (Solagro et al. 2016). This diet, based on the BioNutriNet project, a part of the NutriNet Santé cohort (Kesse-Guyot et al. 2022), was designed to propose a reduction in the consumption of animal proteins in favour of plant proteins, associated to a limited reduction of energetic and proteic intakes. It is used in to evaluate how nutrient deposits might change if the population turned to more plant-based diets.

- Prospective French diets from the ADEME's Transition2050 scenarios and the SISAE project (Barbier et al. 2022), are also derived from the BioNutriNet project. These scenarios propose four consistent yet contrasted pathways for the agricultural and food transition towards climate neutrality in France. The scenarios range from a "frugal generation" (scenario 1 – meat consumption reduced threefold) to a "restorative gamble" (scenario 4 – meat consumption limited to a 10% decrease). Scenario 1, in particular, is used to assess whether further changes in the nutrient deposits might come from a more significant reduction of caloric and proteic intakes in addition to a shift towards more plant-based products.

- The prospective Planetary Health Diet (PHD, Van Rossum et al. 2023) with specific proposal for the Netherlands (Uriza-Pinzón et al. 2024), to match the case study of Amsterdam; this diet was introduced with the aim of fulfilling nutritional requirements worldwide while keeping agricultural production within the safe space of the planetary boundaries.

The nutrients intakes of the different diets used in this article are available in the appendix Table A3.

2.3. Modelling urban metabolism

The goal of the urban metabolism model is to identify the amount of nutrients excreted by the population and the location where they occur. The evaluation of the urban metabolism is obtained through the following steps:

1. the population physically present within the area, over a period of interest, is quantified in terms of the number of residents, workers, students, and tourists;

2. the time spent by each population in the place of residence, work, study, or in other locations is estimated;

3. this information is coupled to the metabolic model and excretion patterns to quantify the amount of nutrient excreted in each location.

A detailed methodological framework for the first two steps can be found in Redlingshofer et al. (2025) and will be briefly explained below. The generic nature of this method allows it to be applied in any kind of are, over any time-frame greater than a day, provided that the relevant demographic and nutritional data is available. In this work, it is used on two case studies: the Toulousian area (France) and Amsterdam (the Netherlands).

2.3.1. Identification of the population structure in the territory

The population spending time in the studied territory is estimated based on demographic data including the age group and the type of activity performed. The gender is not included in the collected data, the population was considered equally split between male and female. For each area of interest, one must obtain the resident population P_r (Equation 5), the commuting population P_c (Equation 6), and the tourist population P_t.

$$\mathrm{P}_r = \sum_g \mathrm{P}_{r,g} \tag{5}$$

$$\begin{aligned} \mathrm{P}_c &= \mathrm{P}_{\text{employees (e)}} + \mathrm{P}_{\text{students (s)}} \\ &= \sum_g \mathrm{P}_{e,g} + \sum_g \mathrm{P}_{s,g} \end{aligned} \tag{6}$$

Each population is composed of different age groups g except for the tourist population for which this information is not available.

In the case of Toulouse, the French national statistical service[3] (INSEE) provides past and current demographic data at different scales, which enables to compute the residents, the employed or unemployed parts of the active population, and the tourists. The French ministries for (higher) education provide data for students. Sources are detailed in the Appendix B.

For Amsterdam, resident, employee, student and tourist-related data were obtained from the CBS Statline open data portal[4].

2.3.2. Coupling population to human metabolism and dietary data

The second step in modelling urban metabolism is to allocate the nutrient flows to the places where they are excreted. The principle is to assign a quantity of excreta and nutrients according to the estimated amount excreted by each of the populations given the local diet.

The working population (P_e) is used to compute the amount of urine, feces, and associated nutrients excreted at work: for each working day, based on the typical schedule of an employee, we estimate the fraction of mass and nutrient excreted during that fraction of the day, cf. Table 1. The same procedure is used for students during their time at school and for residents during their time at home (either on days where they don't commute, or in the morning and evening of working days). The days spent by residents outside of the territory when they leave during holidays and the tourists overnight stays are also taken into account using the framework developed in Redlingshofer et al. (2025).

Based on the literature (Burgio et al. 1991; Heaton et al. 1992), one can assess the fraction of mictions and defecations that occur at work, at home, or in other areas – see Table 1 and Appendix A for details. These excretion patterns give an average excretion at work of around 55% for both urine (estimated range: 49–61%) and feces (46–67%), as detailed in the supplementary notebook section "Individual habits: linking human and urban metabolisms". The archetypal urination patterns can thus be approximated as 1 miction in the morning, before leaving for work, 4 mictions at work, then two in the evening, after work (this would lead to 57% of urination at work on a commute day, very close to the 55% obtained from Burgio et al. (1991)). Based on the simplified urination pattern, the fraction of excretion at the place of study was approximated as 3/7 mictions for students, with 1 miction during each break (in the morning, at noon, and in the afternoon). This reduced fraction also tries to account for their documented preference for excreting at home rather than at school (Lundblad and Hellström 2005). Defecation patterns are more peaked in the morning, with an average slightly above one defecation per day per person (Heaton et al. 1992). Since the peak is close to the typical time where people leave for work, there is a high uncertainty in the precise distribution of defecation between home and work. We therefore chose to use the same fraction for both defecation and urination (55% at work on a commute day). At school, however, a 20% fraction was chosen to account for the fact that 60% of children may avoid defecating there (Lundblad and Hellström 2005), leaving only 40% of children with defecations that are hypothesized as equally split between home and school.

3. https://www.insee.fr
4. https://opendata.cbs.nl

Source	Fraction excreted at work	Fraction excreted at school
Urine	55% (49–61%)	43%†
Feces	55% (46–67%)	20%†

Table 1: Estimated excretion fractions at the place of work or study during a commute day. Fractions at work are directly computed based on daily excretion patterns from Burgio et al. (1991) and Heaton et al. (1992) and average workers' schedules (estimated range is given in parenthesis). Fractions at school are instead estimated based on proposed urination patterns and try to account for reported toilet avoidance by children at school (Lundblad and Hellström 2005). †Uncertainty for excretion at school could not be quantified.

2.4. Case studies

The method has been applied on two case studies: the Toulousian area, in France, and the city of Amsterdam, in the Netherlands. The French area was chosen because our partnership with local actors gave us access to significant amounts of data, including prospective data on the evolution of the population on a territory composed by a variety of municipalities with diverse profiles. Amsterdam was chosen as a complementary case study because multiple publications have investigated its sanitation system and the potential of source separation in this area, providing reference data that can be used to compare the new model's results.

To validate the human/urban metabolism model, the nutrient flows predicted by the model are compared to the nutrient flows at the entrance of wastewater treatment plants (WWTPs) of the area. The difference between the model and the data was used to assess the composition of the greywater and check whether it was compatible with typical values found in the literature (Meinzinger and Oldenburg 2009; Comber et al. 2013; Starck et al. 2024).

2.4.1. The Toulousian area

The investigated area around Toulouse is constituted by 66 municipalities grouped in three large wastewater collection systems and 16 smaller ones. The population of the studied area was 985,387 inhabitants in 2022, the reference year for this study.

Wastewater treatment data comes from 23 facilities treat the wastewater of the area for a total capacity of 1,402,070 population equivalent (PE, defined as 60 gBOD5/day)[5]. The largest WWTP (Ginestous-Garonne) has a capacity of 950,000 PE. Because the multiple collection systems can be interconnected and data about the precise fraction of the population going to each plant was not available, we chose to aggregate the data from all contiguous sanitation areas around Toulouse. The total excretion of the population in this territory was thus compared to the total amounts of nitrogen and phosphorus entering the WWTPs, accounting for loss of wastewater from improper wastewater connection or damage of wastewater pipe or non-reported storm overflow. The sum of these three different losses of wastewater has been already estimated to 7% of the flowrate of wastewater at the scale of the water body Adour–Garonne (Starck et al. 2023).

In addition to the nitrogen and phosphorus loads from excretions, we also included the contribution from greywater, with a nitrogen load of 0.1–1.7 gN/cap/day Meinzinger and Oldenburg (2009) and a phosphorus load of 0.1–0.6 gP/cap/day. For phosphorus, we used the upper bound from Starck et al. (2024) to have a more recent data point from France, but used the lower bound of Meinzinger and Oldenburg (2009) as the lower bound in Starck et al. (2024) was 0 gP/cap/day (due to the uncertainties in the data used for the study), which is not realistic for typical greywater. Nitrogen deposition was

5. https://adour-garonne.eaufrance.fr/catalogue/581d5f70-558c-49e4-8d77-5bd4fe974b62 (SIEAG, archived version)

not taken into consideration in Toulouse since 99% of its sewer is separated from rainwater[6].

For the prospective analysis, we used the estimated projections from the AUAT[7], which give the expected population and employment in the area in 2040, which is the time horizon used for strategic planning in the area. The detailed age groups were obtained by combining this data with the age pyramid for France in 2040 from the French national statistics institute[8]. Estimation of student population was made by applying the 2022 students to residents ratios for each age group to the residents of 2040.

More details about the population and wastewater data can be found in Appendix B and in the supplementary code.

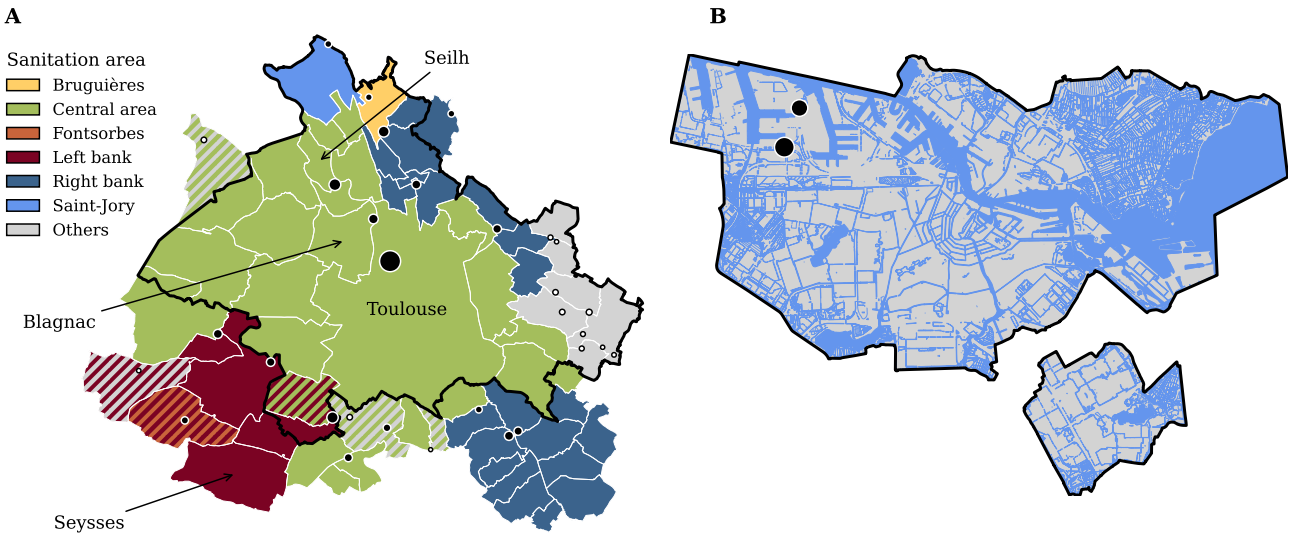


Figure 2: Wastewater treatment plants (WWTPs) in the two case studies. **A**. Organization of sanitation areas in Toulouse: WWTPs of areas collecting more than 25 tN/y are marked by black-filled disks whereas WWTPs in smaller collection areas are marked by white-filled disks. The limits of "Toulouse Métropole", the inter-municipal cooperation area, is marked by a thick black line. Each collection area is given by a specific colour and municipalities served by two collection areas are marked by hashed colouring. **B**. Map of Amsterdam and the two studied WWTPs; water bodies are shown in blue and the urban area is in grey.

2.4.2. The city of Amsterdam

The second study case is the city of Amsterdam (*Gemeente Amsterdam*), with a population of 753,589 residents older than 3 in 2013. The wastewater of Amsterdam is treated in two large WWTPs named Amsterdam West and Westpoort with a treatment capacity of 733,320 PE and 361,000 PE respectively. 70% of the population of Amsterdam is treated in Amsterdam West and 28% in Westpoort, the remaining goes to other, smaller, WWTPs. Some other cities are also treated in these WWTPs, but Amsterdam represented respectively 94% and 77% of the effluents treated in Amsterdam West and Westpoort in 2013 (Fooij 2015). In addition to the nitrogen and phosphorus loads from excretions, we also included the contribution from greywater using data from Meinzinger and Oldenburg (2009) with a nitrogen load of 0.1–1.7 gN/cap/day and a phosphorus load of 0.1–2.2 gP/cap/day. Since part of the city uses combined sewers and nitrogen deposition in the Netherlands[9] is significant, the mass flow also integrates an estimated contribution from this flow to the wastewater entering the WWTPs

6. https://www.eaudetoulousemetropole.fr/service-public/publications/rapports-annuels-chiffres-cles (Toulouse 2024 water report, archived PDF)
7. Agence d'Urbanisme et d'Amménagement de Toulouse: https://www.aua-toulouse.org
8. https://www.insee.fr/fr/outil-interactif/5896897/pyramide.htm#!y=2040&c=0 (INSEE, archived version)
9. https://www.clo.nl/en/indicators/en018919-nitrogen-deposition-1990-2024 (archived version)

(Marra et al. 2025). A more detailed repartition by age for residents, workforce, students, retirees and tourists can be found in A9 in Appendix C and in the supplementary code.

Results from the model are compared to wastewater data and results from previous studies on the city of Amsterdam, namely: Fooij (2015) and Van der Hoek et al. (2016) for nitrogen and phosphorus in 2013, Wielemaker et al. (2020) (mixed data, mostly between 2014 and 2017) and Van der Hoek et al. (2018) for nitrogen in 2017.

For the prospective analysis, we used data from the national Dutch statistics[10] and the Amsterdam statistics council[11] that provides a range for the expected population growth between 2021 (reference year with the last dietary survey) and 2040. The detailed age groups were obtained by combining this data with the age pyramid for the Netherlands in 2040[12]. Estimation of the employed and student populations was made by applying the 2022 students- or workers-to-residents ratios for each age group (using appendix Table A8) to the residents of 2040.

3. Results

3.1. Model for human metabolism

Following the data analysis shown in Figure 1, the final parameters are shown in Table 2. For nitrogen, the choice was made based on the fact that reported losses of nitrogen from sweat and other processes did not seem to exceed 3 to 4% of the total nitrogen intake, so we fixed $a_l^{(N)}$ to 4% and took the upper bounds of the fitted values for urine and feces. For phosphorus, the various losses seem negligible (Johnston et al. 1958; Peacock 2021) compared to urinary and fecal excretion and a loss value of zero was compatible with the fit. Given that the quality of the fit was higher for feces, we fixed $a_f^{(P)}$ to 43% and set $a_u^{(P)} = 1 - a_f^{(P)}$, which was compatible with the result of the orthogonal distance regression.

Compound	**Fraction in urine**	**Fraction in feces**	**Fraction in other losses**
Nitrogen	$a_u^{(N)} = 0.79$	$a_f^{(N)} = 0.17$	$a_l^{(N)} = 0.04$
Phosphorus	$a_u^{(P)} = 0.57$	$a_f^{(P)} = 0.43$	$a_l^{(P)} = 0$

Table 2: Values of the parameters in the final metabolic model. For children, the fractions in urine are smaller and vary with age due to nutrient retention in muscles and bones during growth.

The precise numbers used to study the area around Toulouse (France), and in Amsterdam (Netherlands), were derived from the model and the local diets and are shown on Figure 3A. For each age group, these values were modulated by the fraction of the reference intake that is ingested at that age, as shown on Figure 3B. To compute the volumes of urine excreted, the curve shown in Figure 3C can be used.

3.2. Confronting the model's results to sanitation data

3.2.1. Amsterdam, The Netherlands

In 2013, our model estimates that the excreta reaching the two WWTPs amounted to 3,400 tN and 402 tP, with urine contributing to 2,796 tN and 228 tP. The excreta represented respectively around 82% and 68% of the nitrogen and phosphorus entering the WWTPs through wastewater coming

10. https://www.cbs.nl/en-gb/visualisations/dashboard-population/population-dynamics/population-in-the-future (CBS, (archived version))
11. https://onderzoek.amsterdam.nl/publicatie/de-staat-van-de-stad-amsterdam-xiii-2024-2025 (archived report)
12. https://www.populationpyramid.net/netherlands/2040/ (archived version)

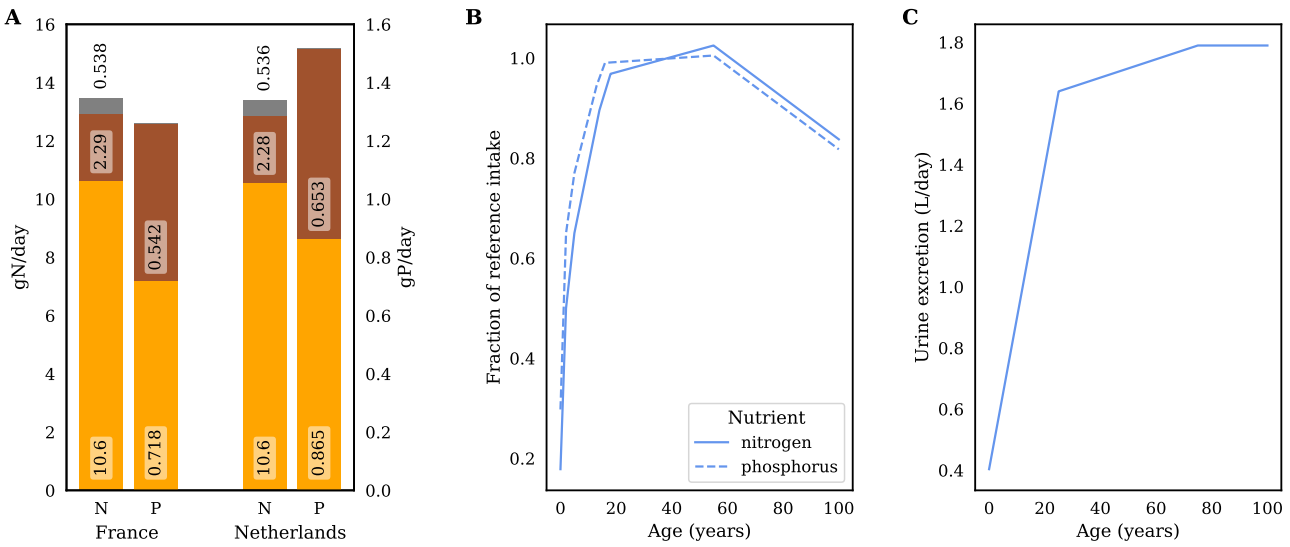


Figure 3: Resulting model for human metabolism. **A**. Proposed values for the distribution of nitrogen (N) and phosphorus (P) between urine (orange), feces (brown), and other losses (grey) in adults, given the diets reported for France and the Netherlands. **B**. Modelled evolution of the fraction of the reference intake for nitrogen (solid) and phosphorus (dashed) that is ingested depending on the age of the individual. Right: modelled evolution of the daily volume of urine excreted with the age of the individual.

from Amsterdam. Greywater was estimated to contribute 26 to 437 tN and 131 to 210 tP. Finally, atmospheric nitrogen deposition via combined sewer was estimated at 74–151 tN. The reconstructed wastewater accounts for 85 to 96% of the nitrogen inflow to sanitation and 90 to 103% of the phosphorus reported in Fooij (2015). Van der Hoek et al. (2016) also estimated a relative contribution of 174.5 tons of phosphorus from business-related buildings in 2013 compared to 357.5 tP from households in the entire area connected to the WWTPs. This gives a 2.05 ratio, which is similar to the 2.17 ratio we obtain through our model, which uses an upper estimate for the residential time (including leisure time that may occur away from home).

We also compared our results to the data from Wielemaker et al. (2020) who used detailed data from the city of Amsterdam to estimate an average excretion of 812 kgP per neighbourhood in 2015, leading to a total excretion of 390.5 tP over the 481 neighbourhoods of Amsterdam. However, the paper also mentions the "citys annual load of 330.5 tons" of phosphorus and we could not confirm whether this figure was a typographic error or the correct value. For this same year, our model predicted 371 tP excreted in Amsterdam, which is compatible with their range of value.

3.2.2. Toulousian area, France

In the Toulousian area, according to the model, urine and feces reaching the wastewater treatment plants (WWTPs) represented 4,015 tN and 393 tP in 2022. In comparison 4,323 tN/year and 475 tP/year are supposed to enter these WWTPs, from the monitoring data, which means excretion is responsible for around 93% of the nitrogen mass flow and 83% of the phosphorus. Greywater contributed to approximately 29 to 488 tN/year and 29 to 172 tP/year. As shown in Figure 4A, the combined entries expected from excreta and greywater cover the entire inflow of the sanitation system.

The contributions of the different municipalities in the area vary by several orders of magnitude, with Toulouse contributing to 54% of the total deposit (2445 tN/year) while Blagnac, with the airport and the aerospace employment areas, represents only 3.6% (164 tN/year) and smaller towns close to rural areas contribute less than 1%. Besides the variations in the amount of excreta generated, different municipalities also display different metabolic profiles regarding the type of population and locations where urine and feces are excreted.

3.3. Local metabolic profiles and spatial analysis

Through the combination of the human and urban metabolic models, the metabolic profiles of different areas can be compared. The relative importance of residents, workers, students or tourists in terms of the amount of nutrients they generate can vary significantly across municipalities within the Toulousian area, as shown on Figure 4B and in appendix Figure A1. Thus, municipalities that contain a large fraction of industrial and tertiary buildings, such as Blagnac (where the airport of Toulouse is located), display a lower-than-average contribution from residents and a larger-than-average contribution from workers and tourists. Conversely, municipalities that are mostly residential and where residents leave to work in neighbouring, more attractive areas, such as Seysses, have a larger-than-average contribution from residents and a smaller-than-average contribution from workers and students. Finally, Toulouse hosts most schools and universities and collects a higher fraction of the nutrients from students compared to the other areas.

Similarly to the work done by Wielemaker et al. (2020), the model can also be coupled to geospatial data or statistical data about the distribution of households and workers within buildings to identify "hotspots" that could be prioritized to recover large amount of nutrients while transforming only a few buildings. Figure 4C shows how, even in small cities such as Seilh (which has around 3,300 inhabitants and represents less than 1% of the total deposit of the Toulousian area), a few collective buildings can produce several hundreds of kilograms of nitrogen every year. Over the whole metropolis, the 125 residential buildings hosting 100 households or more could thus provide 125 tN/year, with the largest 41 buildings producing more than 1 tN/year/building.

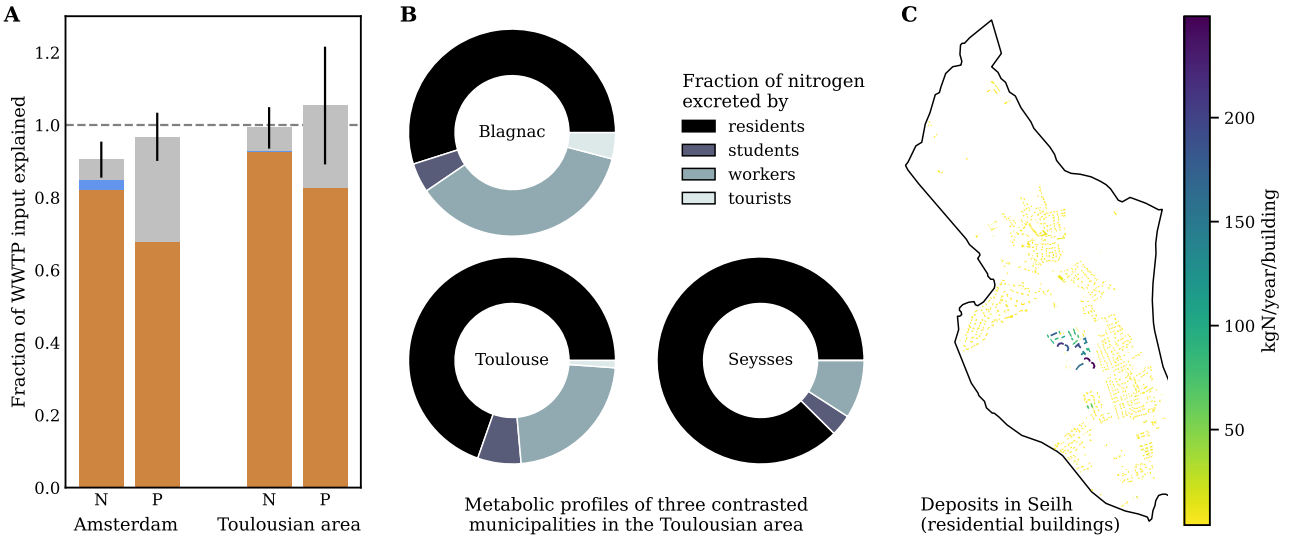


Figure 4: Visualization of the model data for Amsterdam, the Toulousian area and three municipalities within it. **A**. Comparison between the model's prediction of what should enter the local WWTPs and what is actually reported for nitrogen (N) and phosphorus (P). The areas in orange-brown display the contributions from excreta. Grey areas display the expected contribution from greywater, with associated uncertainty. The blue area for nitrogen in Amsterdam estimates the contribution from atmospheric deposition entering the combined sewers; it is absent for Toulouse which has separative sewers. **B**. Metabolic profiles of three municipalities in the Toulousian area displaying different characteristics: Blagnac shows a higher relative contribution from workers and tourists compared to Toulouse, which shows a higher fraction coming from students; the third municipality, Seysses, displays a larger contribution from residents than the other two areas. **C**. Distribution of nitrogen deposits within the residential buildings of the Seilh municipality, displaying significant discrepancies between individual houses (in yellow) and collective buildings (from small ones, in green to large ones in dark violet).

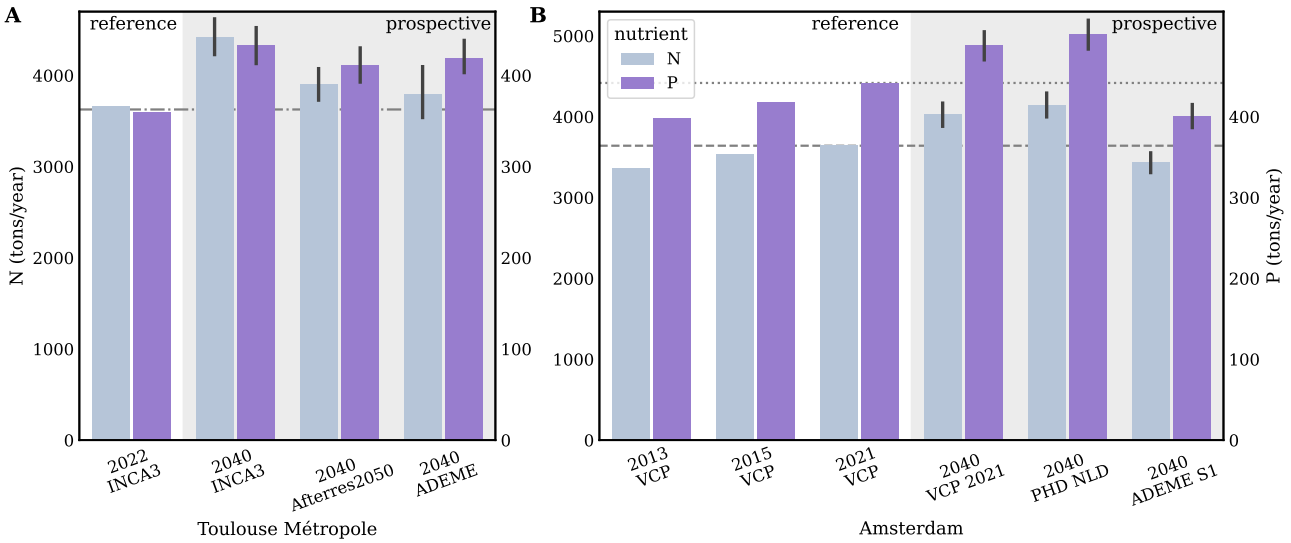


Figure 5: Potential evolution of the nutrient deposits in 2040 in Toulouse Métropole and Amsterdam. **A**. In Toulouse, with a 20% population increase expected between 2022 and 2040, the prospective data shows that, if the diets proposed to decrease the environmental impact of food consumption were broadly adopted by 2040, they could limit (Afterres 2050) or even compensate (ADEME S1, the lowest bound within all ADEME scenarios) the load from the growing population on the sanitation system. **B**. In Amsterdam, that underwent a significant population growth between 2015 and 2021, the reduction of N and P in the diet contributed to limiting the load on the sanitation system. With a mean growth of 13% until 2040, this load could however keep increasing, especially if the diets stay high in N and P, like in the proposed Dutch version of the Planetary Health Diet (PHD). However, as for Toulouse, more environmentally-conscious diets such as the ADEME S1 could compensate the population increase and bring the load below 2021 levels.

3.4. Impact of prospective diets

The potential evolution of the load entering sanitation system was evaluated for the Toulousian area and Amsterdam based on expected population growth and diets, as shown on Figure 5. Note that these are hypothetical scenarios based on prospective population and diets: data for 2040 reflects the possible outcomes of a modelling exercise describing a potential societal change (or status quo).

In both Toulouse and Amsterdam, significant population growth is expected (respectively 20 and 13% on average, cf. Tables on prospective populations in the Appendix), which will increase the inputs to the WWTPs accordingly. However, we show that transitions to more sustainable diets could compensate the population growth and keep the loads stable. The fact that the proposed version of the Planetary Health Diet for the Netherlands (Uriza-Pinzón et al. 2024) leads to higher nitrogen and phosphorus intakes compared to the latest survey of Dutch diets (VCP 2019–2021), as shown on Figure 5B, can be surprising but does not actually contradict its aim of reducing the environmental impact of food consumption. This is detailed at the end of the Discussion's first subsection (4.1).

4. Discussion

4.1. Contribution to the research domain

In this work, we propose a new metabolic model, based on a large dataset of studies on human physiology, that enables researchers and local actors to assess the amount of nitrogen and phosphorus generated by the different populations in their areas through urinary and fecal excretion. This model can be coupled to local data about the diet of the population in a given area at a given time, and is able to account for age and gender differences. Compared to previous works, that used average number

from sanitation or medical measurements on specific populations, the combination of the metabolic model with data from local diets and populations can thus provide results that are better tailored to the situation or area of interest. Open-source Python libraries and example codes are provided to reproduce or adapt these calculations (Appendix, section D) and simplified reference values for different populations are also given in the Appendix, section A.

Combining the metabolic model with statistical data about the present population enables us to extend the work from Redlingshofer et al. (2025) and provide a detailed overview of the downstream contribution of human food consumption to the local urban (or, more generally, territorial) metabolism. This urban metabolic model enabled us to explain the sources of the nutrients entering the wastewater treatment plants (WWTPs) in the vicinity of Toulouse and Amsterdam and to validate the robustness of the model based on sanitation data. The model can also be used to generalize detailed analysis at multiple scales, from territorial analysis (see appendix Figure A1) to building-level analysis by using publicly available data. For instance, this enabled us to reproduce, for the Toulousian area (Figure 4C), the same type of analysis done by Wielemaker et al. (2020) at the buildings' scale in Amsterdam. These results can assist local actors assess which building they should renovate first to separate human excreta from greywater and recover the nutrients for agricultural purposes, depending on the amount of fertilizer they want to produce, the location of the deposits, and the type of buildings they would like to target.

Finally, we show that this model enables to explore prospective scenarios for the sanitation systems. Through them, one can see both the contributions from increasing populations in metropolitan areas and the impact of dietary changes. For instance, regarding nitrogen, we show that the reduction of nitrogen in diets (from reduced caloric intake, reduced consumption of animal products, or both) could limit the increase of the nitrogen load entering the sanitation systems despite rising populations. Thus the ADEME S1 diet, featuring both reduced consumption of animal-based proteins and reduced caloric intake, could more than compensate the increase in population and enable a reduction of the nitrogen load if they were broadly adopted.

However, not all prospective diets lead to reduction in nutrients' intake: the proposed declination of the Planetary Health Diet (PHD) for the Netherlands, for instance, displays higher nitrogen and phosphorus intakes than the current Dutch diet. Though this might seem surprising for as the PHD aimed at reducing the environmental impacts of food consumption, it can be explained by two main factors. First, the diet was proposed as a comparison to the Dutch dietary guidelines, which advise even higher nutrient intakes, and not to current diets. Second, the fact that they lead to higher nutrients' intake should not be mistaken for higher environmental impact: since the latter is mostly dominated by the source of the food (plant or animal-based) and the resources necessary to produce it (Crippa et al. 2021), the PHD can have lower environmental impact (accounting for all steps between food production and excreta management) while generating higher N/P loads at the WWTPs. It should also be noted that the proposal from Uriza-Pinzón et al. (2024) is one specific declination of the first version of the PHD and alternative proposals, especially if based on the following version (Rockström et al. 2025), may lead to different nitrogen and phosphorus intakes.

For phosphorus, the results depend on the current P diet in the area. In the Netherlands, where people have a high-P diet the load could also decrease but in the Toulousian area, where people have a lower P consumption, dietary changes are unlikely to compensate the expected population increase.

This is of significant relevance for cities, as many have to perform major investments in WWTP upgrades to comply with the revised *EU Directive 2024/3019* (2024) on urban wastewater. As such, any reduction in the nutrient loads entering the sanitation system, be it from dietary changes or source separation projects, could lower the investments required to comply with the new requirements or even help avoid major upgrades altogether (as discussed in *Étude sur la structuration de filières de séparation à la source des urines en région Ile-de-France* (2026) — in French).

4.2. Comparison to the literature and specific case studies

Comparing the flows of nitrogen and phosphorus that the model predicts with the values reported by the WWTPs of the Toulousian area, we were able to explain the integrality of the nutrients entering the facilities. Compared to previous studies done on the French sanitation system (Starck et al. 2023), we obtain the same contribution (93%) from excreta to the nitrogen load of the sanitation system. The proportion of phosphorus coming from excreta that we obtain in the Toulousian area (83%) is higher than the value reported by Starck et al. (2024) at the scale of France (73%) and of the water basin Adour–Garonne (68%) but is well within the uncertainty range estimated in the study (71–94%). Furthermore, phosphorus loads from greywater has significant uncertainty, partly due to a decrease of phosphorus content in detergent over the past years.

In the case of Amsterdam, the known sources could not explain the total amount of nitrogen entering the wastewater treatment plants (WWTPs), leaving at least 4% of the input (around 160 tN) unaccounted for, which may be explained by several factors. First, the amount of tourists for 2013 is most probably underestimated, with a report from Van Benthem et al. (2017) mentioning around 11.5 million overnight stays and a possible contribution from non-reported sources (notably AirBnB) estimated between 1 and 2 million overnight stays. Taken together, these could explain around 50–60 tN, that is 1–1.5% of the nitrogen entering the WWTPs. Second, as the city of Amsterdam is the most attractive area of the metropolis, there is probably a significant fraction of the residents within the urban area that come to the city centre for leisure on evenings or week-ends. This "excursionist" behaviour could increasing the load on the sanitation system of the city centre, as was observed for food consumption in Paris (Barles 2009; Redlingshofer et al. 2025), where it led to a very significant increase (40%) and could thus explain a few percentage points in the case of Amsterdam.

Compared to previous studies, the metabolic model, coupled to data from local diets, leads to substantial differences, either in the amount of nutrient excreted in the area or in the distribution of those nutrients between urine and feces. Both points are significant for operational planning since they will directly affect the downstream value chains in terms of quantity, type, and quality of the fertilizers or soil conditioners that could be produced from the raw material. Thus, Wielemaker et al. (2020) considered an excretion of 1 gP/day from urine and 0.5 gP from feces; while the total is very close to the values for adults reported by the food consumption surveys that we used, they expect that 67% of the phosphorus is recoverable from urine while our model suggests that the ratio in urine is probably lower than 61%, which could affect the decision-making process and might, for instance, make some projects switch from a focus on urine recovery to solutions dealing with blackwater (urine and feces) if they want to recover more phosphorus. Similarly, Van der Hoek et al. (2018) considered an average excretion for the whole population of 1.4 gN/cap/day in feces whereas our model estimate the likely range to be 2.09–2.18 gN/cap/day in the Netherlands, at that period.

4.3. Uncertainties, limitations and perspectives

As mentioned previously, one of the main sources of uncertainty when analysing territorial metabolism is the impossibility of precisely quantifying the present population and its excretion patterns. Though the methods from Redlingshofer et al. (2025) try to account for all available data, they cannot account for excursionists, who may have a significant contribution near the centre of attractive cities, and tend to underestimate the number of tourists. Though this could later be included, the model also does not account for remote work at the moment. Another source of uncertainty is the variability of the dietary intake throughout different regions of a country, as the model currently uses an average value. Moreover, the model only considers the body mass change for children and teenagers under 18 years old, though other categories of people encounter mass change, e.g. elders above 70 years old or pregnant women, that should lead to a change in the nutrient excretion rate. The combined uncertainties from the population and its dietary intake are unfortunately quite difficult to quantify and we chose not to include them in this work.

Another source of uncertainty is the defecation habits outside household. In this work we consider

the same ratio of excretion for urine and feces at work (55% for a commute day). Unlike with children at school, who avoid going to the toilets, especially for defecation, there are few studies on this subject for adults.Wielemaker et al. (2020) used a ratio of 14% of urinations and 4% of defecations occurring in the place of work, based on figures from a 1998 report for the Netherlands, meaning there would be one defecation for 20 urinations there. However, a survey carried out in an office of Toulouse (Besson 2020) obtained a value of one defecation for 4 urinations in the workplace. Since our estimation based on physiological excretion patterns during the day (Burgio et al. 1991; Heaton et al. 1992) gave us one defecation for 7 urinations, we decided to opt for this value; however, this highlights the need for more data to better constrain urination and defecation patterns between the places of work, study and residence.

When comparing the phosphorus load to sanitation data, the phosphorus content in greywater was difficult to assess reliably due to its evolution after the limitations applied to laundry detergent since 2013, and to dishwasher detergents since 2017 (EU Regulation 259 2012). For instance, in Toulouse, the phosphorus load in wastewater has remained constant since 2000 whereas the nitrogen load increased by a factor of 1.45 due to the population increase[5]. More generally, the reliability of the measurements from the WWTPs are also difficult to assess but are assumed to be around 10% for the largest stations' yearly average and larger for smaller stations (Starck et al. 2023).

While the urban metabolism model developed in this study is a first step towards decision-support for source-separation planning, it would require additional developments to provide all necessary information (e.g. variations in nutrient loads reaching a new process unit during the day, volumes to be treated, etc). To that end, both spatial and temporal precisions could be refined in future works. Regarding spatial accuracy, it could be relevant to break down the excretion that occurs neither at home nor at work, for example during leisure activities. In this study, these have been included under home excretion. According to the survey on the daily habits of the French population[13], this time spent "away from home" accounts for 5% of the total. It could be also interesting to investigate the production in the different commercial and institutional building as it has been performed in the work of Crane et al. (2024) based on US habits. Regarding the temporal precision, while the model can be used to compute the deposits over any time-period greater than a day (provided the information on population presence can be obtained, cf. appendix Figure A2A), intra-day variations are not available. Though such data is already taken into account, in an aggregated form, to obtain the spatial patterns (e.g. excretion at work/school and at home), the exact time of excretion is not yet integrated in the model.

The human metabolic model could also be improved in several ways. First, the estimation of nutrient ingestion depending on the age was performed based on data for the US and France and might not match growth curves and food consumption in non-Western countries. Re-fitting the model on data from other parts of the world would be useful to assess the validity of the model for children and adolescents there; changing the model to use nutrient intake by kilogram of body mass could also be a way to make the model more generic, though this would be at the cost of more complexity for the end-user that would need to find and enter the specific weight values for different age and gender groups. Another limitation of the model is that it does not account for variation in the nutrient content of urine throughout the day as we could not find enough data to model it. However these variations do exist (the first mictions of the morning are more concentrated that later mictions) and should be taken into account in future models. Finally, other nutrients, notably potassium, an essential nutrient in agriculture, could be added to the metabolic model. Potassium was not included in this work because it is lost in large quantities through perspiration, which varies greatly depending on individual habits and weather (Consolazio et al. 1963; Malhotra et al. 1976). Furthermore, as is not measured by WWTPs, a potassium model cannot be validated the way nitrogen and phosphorus were. Additional work would thus be necessary to provide and validate a similar model for potassium, which would also need to be further parametrized, when used, to account for typical sweating patterns and meteorological conditions.

13. https://www.insee.fr/fr/statistiques/1281050?sommaire=2118074 (INSEE, archived version)

5. Conclusion

In this study, we provide a method to assess nutrient flows in the biometabolism of human settlements by combining a model of human metabolism, based on medical data and dietary surveys, with statistical data on the population that is present in the area of interest. The results from the model were validated against data from wastewater treatment plants in France and the Netherlands, then used to assess the potential evolution of the nitrogen and phosphorus loads entering the sanitation systems by 2040. This highlighted the potential impact of plant-based diets to reduce nitrogen load, potentially compensating the increasing population, even in attractive cities like Toulouse and Amsterdam. The model is generic and provided through open-source Python libraries so that it can be reused and adapted to assess local deposits in multiple areas throughout the world. Combined with geospatial data on work, education, and residential buildings, the model enables to precisely localize the deposits and thus support decision-making on local source separation projects to secure their fertilizer supply chain and improve food security and autonomy in the area.

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Appendix

A. Sources and data for the human metabolic model

A.1. Nutrient retention in children

The data and analyses performed to estimate nutrient retention in children are available in a Jupyter notebook within the `OrgMatt` library: https://git.sr.ht/~tfardet/OrgMatt/tree/v1.0.0/item/extra/extracted_data/nutrient_retention.ipynb. The 12 medical studies used are present in the `body_composition.csv` file in the library's data folder: https://git.sr.ht/~tfardet/OrgMatt/tree/v1.0.0/item/orgmatt/data/

A.2. Compiled excretion data

	Nitrogen				Phosphorus			
	Age categories and data for France							
Age	**Urine**	**Feces**	**Loss**	**Retention**	**Urine**	**Feces**	**Loss**	**Retention**
0–3	77.09%	17%	4%	1.91%	51.77%	43%	0%	5.23%
4–10	77.13%	17%	4%	1.87%	51.85%	43%	0%	5.15%
11–18	76.71%	17%	4%	2.29%	50.71%	43%	0%	6.29%
18–24	78.69%	17%	4%	0.31%	56.16%	43%	0%	0.84%
25–64	79%	17%	4%	0%	57%	43%	0%	0%
≥ 65	79%	17%	4%	0%	57%	43%	0%	0%
	Age categories and data for the Netherlands							
Age	**Urine**	**Feces**	**Loss**	**Retention**	**Urine**	**Feces**	**Loss**	**Retention**
0–3	77.09%	17%	4%	1.91%	51.77%	43%	0%	5.23%
4–12	76.95%	17%	4%	2.05%	51.37%	43%	0%	5.63%
13–18	77.17%	17%	4%	1.83%	51.97%	43%	0%	5.03%
19–24	78.79%	17%	4%	0.21%	56.41%	43%	0%	0.59%
25–64	79.00%	17%	4%	0.00%	57.00%	43%	0%	0.00%
≥ 65	79.00%	17%	4%	0.00%	57.00%	43%	0%	0.00%

Table A1: Fraction of the nitrogen and phosphorus intake excreted in urine or feces, retained in the body, or lost through sweat, hair loss...

	Nitrogen (gN/cap/day)					Phosphorus (gP/cap/day)				
Age categories and data for France										
Age	**Intake**	**Urine**	**Feces**	**Loss**	**Ret.**	**Intake**	**Urine**	**Feces**	**Loss**	**Ret.**
0–3	5.6	4.3	1.0	0.2	0.1	0.7	0.37	0.31	0	0.04
4–10	9.5	7.3	1.6	0.4	0.2	1.0	0.53	0.44	0	0.05
11–18	12.2	9.3	2.1	0.5	0.3	1.2	0.62	0.53	0	0.08
18–24	13.1	10.3	2.2	0.5	0	1.3	0.71	0.54	0	0.01
25–64	13.6	10.7	2.3	0.5	0	1.3	0.72	0.55	0	0
≥ 65	12.0	9.4	2.0	0.5	0	1.1	0.63	0.47	0	0
Age categories and data for the Netherlands										
Age	**Intake**	**Urine**	**Feces**	**Loss**	**Ret.**	**Intake**	**Urine**	**Feces**	**Loss**	**Ret.**
0–3	5.6	4.3	1.0	0.2	0.1	0.8	0.43	0.36	0	0.04
4–12	9.7	7.5	1.7	0.4	0.2	1.3	0.65	0.54	0	0.07
13–18	12.3	9.5	2.1	0.5	0.2	1.5	0.77	0.64	0	0.07
19–24	13.0	10.2	2.2	0.5	0	1.5	0.84	0.64	0	0.01
25–64	13.4	10.6	2.3	0.5	0	1.5	0.86	0.65	0	0
≥ 65	11.8	9.4	2.0	0.5	0	1.3	0.75	0.56	0	0

Table A2: Intake and excretion of nutrient for the different age groups. Ret. = retention.

A.3. Dietary intakes

Diet name	Energy (kcal/day)	N (gN/day)	P (mgP/day)	Source
INCA3	2159	13.46	1260	ANSES (2017)
Afterres2050	2095	11.87	1198	Solagro et al. (2016)
ADEME S1: frugal generation	1609	10.86	1218	ADEME (2021)
ADEME S2: regional cooperation	1620	11.17	1199	ADEME (2021)
ADEME S3: green technologies	1639	11.32	1176	ADEME (2021)
ADEME S4: restoration gamble	1784	12.83	1285	ADEME (2021)
Dutch diet: VCP 2012–2016	2232	12.86	1458	Van Rossum et al. (2020)
Dutch diet: VCP 2019–2021	2100	12.74	1482	Van Rossum et al. (2023)
Planetary Health Diet (NLD)	1952	13.1	1524	Uriza-Pinzón et al. (2024)

Table A3: Amounts of energy, nitrogen and phosphorus in the reference adult intake of different diets. Reference values for the INCA3 and VCP 2019–2021 diets, that come from surveys, are the year-span weighted mean of the 18–64 age group. Reference values for the VCP 2012–2016 diet are the year-span weighted mean of the 19–70 age group.

A.4. Excretion patterns between work and home

The distribution of urinations and defecations between home and work on a commute day were estimated from Burgio et al. (1991) and Heaton et al. (1992) based on the hypothesis of an 8-hour workday, with 1 to 2 hours of break and a commute time between 15 minutes and 1 hour. Considering that the one-way commute time is associated to excretion at the place of work and that people typically leave home between 7:30 and 8:30 AM, this gave an average distribution of 55% of urine and 57% or feces excreted at the workplace. Given the limited difference between the two, we thus chose to consider 55% for both.

Detailed calculation are available in the supplementary code, see section D.

B. Data for Toulouse Métropole

The source used to compute the populations are detailed in Table A4, the resulting population in the Toulousian area and Toulouse Métropole are given in Tables A5 and A6 respectively. The data for the sanitation system where obtained from the local water agency's website: https://adour-garonne.eaufrance.fr/usages-et-pressions/polluantes-domestiques-et-urbaines.

INSEE codes of the municipalities in the Toulousian area, with the codes of those belonging to Toulouse Métropole first, in bold: **31003, 31022, 31032, 31044, 31053, 31056, 31069, 31088,**

31091, 31116, 31149, 31150, 31157, 31163, 31182, 31184, 31186, 31205, 31230, 31282, 31293, 31351, 31352, 31355, 31389, 31417, 31418, 31445, 31467, 31488, 31490, 31506, 31541, 31555, 31557, 31561, 31588, 31160, 31188, 31057, 31291, 31460, 31162, 31161, 31575, 31424, 31547, 31402, 31409, 31036, 31254, 31113, 31458, 31035, 31384, 31169, 31446, 31429, 31284, 31433, 31411, 31203, 31187, 31526, 31410, 31420.

Population	**Source**
Residents	French national census: https://www.insee.fr/fr/statistiques/1893204
Commuters	Work commute survey: https://www.insee.fr/fr/statistiques/7630376 Study commute survey: https://www.insee.fr/fr/statistiques/7630372
Tourists	Overnight stays: https://www.insee.fr/fr/statistiques/series/116147542?INDICATEUR=2414593 (see Redlingshofer et al. 2025 for details)

Table A4: Sources used to compute the present population in the Toulousian area.

age	**group**	**residents**	**workforce**	**students**	**retirees**	**tourists**
3–5	kid	32,071	0	32,071	0	0
6–10	kid	54,460	0	54,460	0	0
11–14	teenager	42,648	0	47,988	0	0
15–17	teenager	31,986	0	28,075	0	0
18–24	young adult	110,748	0	143,924	0	0
25–64	adult	460,236	570,611	0	0	1,579,841
≥ 65	senior	231,793	0	0	231,793	1,579,841
Total		985,387	570,611	327,963	231,793	3,159,682

Table A5: Population in the Toulousian area for the year 2022 (source: INSEE), used for comparison with the sanitation data on Figure 4A.

age	group	residents	workforce	students	retirees
< 3	toddler	17,240	0	17,240	0
3–5	kid	26,213	0	26,213	0
6–10	kid	43,979	0	43,979	0
11–14	teenager	33,926	0	39,471	0
15–17	teenager	25,445	0	26,963	0
18–24	young adult	100,269	0	134,128	0
25–64	adult	382,202	509,071	0	0
≥ 65	senior	231,793	0	0	189,217
Total		818,491	509,071	287,993	189,217

Table A6: Population in Toulouse Métropole for the year 2022 (source: INSEE), used for comparison with the prospective scenarios in Figure 5.

age	group	residents	workforce	students	retirees
≤ 3	toddler	40,791 (34,542–47,562)	0	40,791 (34,542–47,562)	0
4–10	kid	60,962 (52,126–70,362)	0	60,962 (52,126–70,362)	0
11–17	teenager	83,425 (77,703–89,173)	0	93,350 (86,947–99,781)	0
18–25	young adult	86,952 (84,932–88,967)	0	116,313 (113,612–119,009)	0
26–64	adult	445,001 (434,697–455,102)	638,122 (609,795–667,748)	0	0
≥ 65	senior	258,093 (247,933–269,336)	0	0	258,093 (247,933–269,336)
Total		975,225 (931,933–1,020,501)	638,122 (609,795–667,748)	311,417 (287,227–336,713)	258,093 (247,933–269,336)

Table A7: Prospective population for Toulouse Métropole in 2040 (source: AUA Toulouse), used for the prospective scenarios in Figure 5. First number gives central hypothesis while the range in parentheses gives the lower and upper estimates.

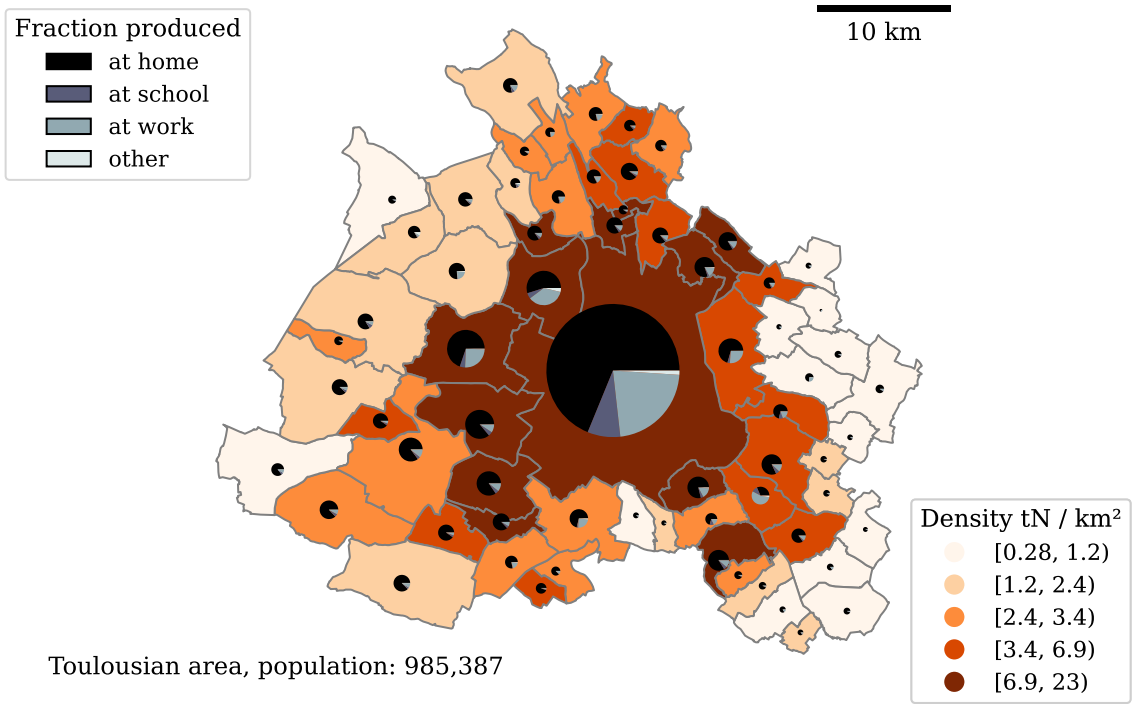


Figure A1: Density of the deposit and distribution between residential, work, education and touristic places over the municipalities of the Toulousian area

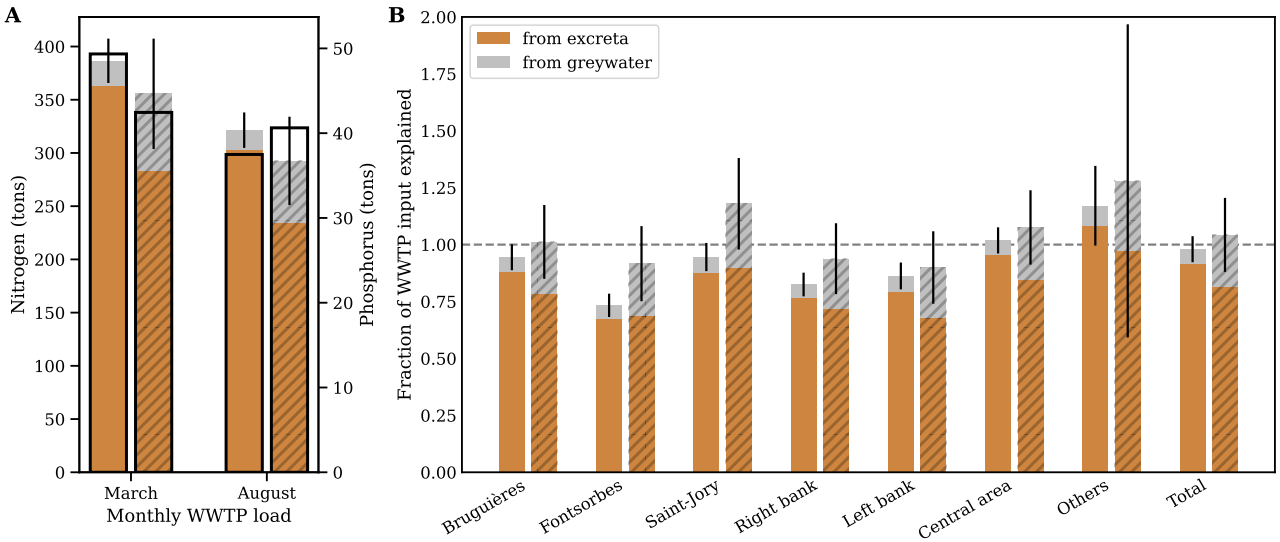


Figure A2: Detailed comparison by period of the year (A) or wastewater collection area (B) in the Toulousian area in 2022. Plain bars are for nitrogen (N), hashed bars for phosphorus (P). **A**. Comparison between the absolute measured (black bars) and estimated flows entering the stations in March and August. **B**. Comparison of the estimated fractions of N and P entering stations from excreta and greywater.

Figure A1 shows that the deposits are mainly concentrated around Toulouse, with a few cities like Blagnac or Labège showing large deposits in the workplace while cities in the periphery are mostly residential.

Figure A2A shows how the model can be used at timescales smaller than a year, provided the relevant data on the populations are available. In France, touristic data is available down to months, and we estimated 3 days of holiday, 21 working days, and 19 studying days for March 2022 against 5 days of holiday, 18 working days, and no studying days in August 2022. We also considered that higher education students were absent during this month. Changes in the N/P ratio may partly be due to changes in the seasonal diet, and partly from the normal variability in the measurements. Figure A2B illustrates the variability of the wastewater measurements when considered at spatial scales smaller than the whole area. It is difficult to assess whether discrepancies between the model and the data stem from actual phenomena that were not accounted for or from the accumulation of imprecise delimitations, unregistered transfers between collection systems and imprecise measurements.

C. Data for the city of Amsterdam

Average nitrogen deposition values in the Netherlands vary quite significantly depending on the source. According to Dutch official figures[14], the average deposition in 2013 was between 1497 and 1637 molN/ha, i.e. 21–23 kgN/ha. In contrast, the atmospheric chemistry and climate model intercomparison project (ACCMIP) and the FAOStat Cropland Nutrient Balance dataset estimate values between 11.2 and 12.8 kgN/ha during this period. The city's area is 219.32 kmš in total, with 30% connected to combined sewers. The fraction of atmospheric deposition entering the WWTPs through rain events should thus be between 74 and 151 tN.

14. https://www.clo.nl/en/indicators/en018919-nitrogen-deposition-1990-2024

age	group	residents	workforce	students	retirees	tourists
4-12	kid	72,204	0	62,420	0	0
13-18	teenager	41,998	0	37,550	0	0
19-24	young adult	72,760	0	75,090	0	0
25-64	adult	468,847	554,100	0	0	4,147,919
≥ 65	senior	97,780	0	0	97,780	4,147,919
Total		753,589	554,100	175,060	97,780	8,295,838

Table A8: Population by age in Amsterdam for the year 2013 (source: CBS), used for comparison with the sanitation data on Figure 4A.

age	group	residents	workforce	students	retirees	tourists
4-12	kid	74,056	0	63,570	0	0
13-18	teenager	43,637	0	38,920	0	0
19-24	young adult	74,292	0	73,900	0	0
25-64	adult	498,079	579,500	0	0	4,958,858
≥ 65	senior	102,725	0	0	102,725	4,958,858
Total		792,787	579,500	176,390	102,725	9,917,716

Table A9: Population by age in Amsterdam for the year 2015 (source: CBS)

D. Supplementary code and data

All the code and data used to compute the results and figures in the article is available on the following git repository: https://codeberg.org/tfardet/article_human_urban_metabolism/src/branch/main/code_article.

The `README.md` file explains how to install the required software and libraries to run the code.

The main libraries used are:

`OrgMatt` https://orgmatt.readthedocs.io/

`gistools` https://git.sr.ht/~tfardet/gistools

`pycafe` https://git.sr.ht/~tfardet/pycafe

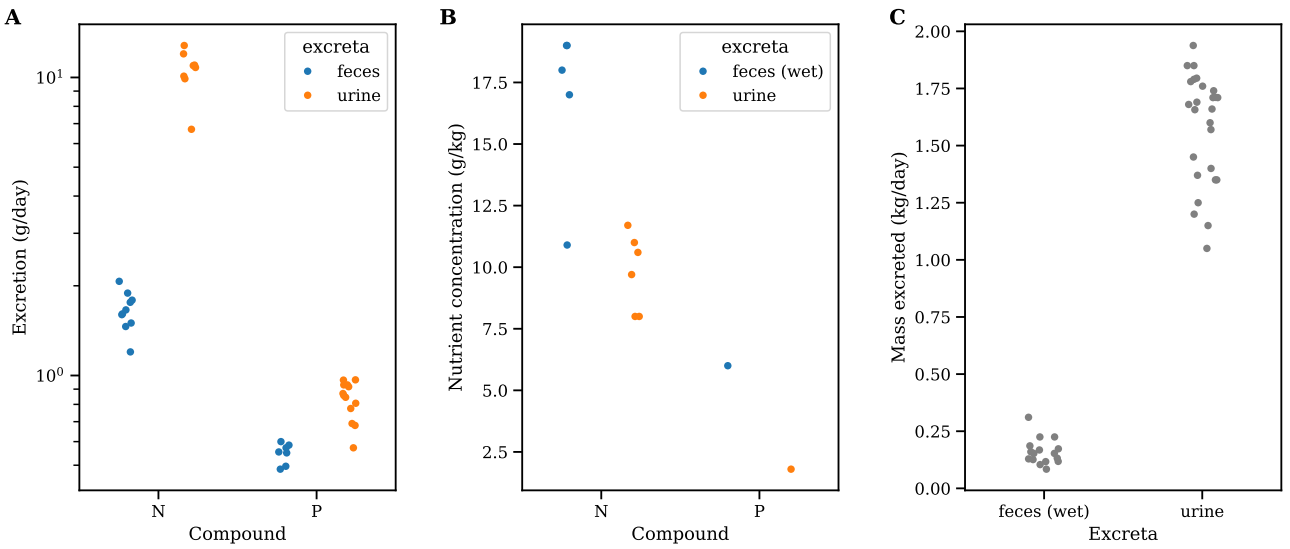


Figure A3: Variability of the average excretions values reported for adults in the medical literature. **A**. Variability of average daily nutrient excretion in urine and feces. **B**. Variability of the average nutrient concentration in urine and feces (insufficient data was found for phosphorus). **C**. Variability of the amount of urine and feces excreted, on average, during a day.

D.1. Data on excretion variability

Figure A3 shows the variability of the average values reported in the literature regarding human excretion. The precise list of values is available in the dataset of the `OrgMatt` library, together with the references of the associated studies, in the following CSV files:

- `excretion_content.csv`
- `excretions.csv`
- `nutrients_flows_body.csv`

D.2. Code and data for the human metabolism

The data used to perform the linear regressions for the adult part of the metabolic model is gathered in the following CSV file: https://codeberg.org/tfardet/article_human_urban_metabolism/src/branch/main/code_article/metabolism/nutrient_intake_vs_output.csv.

Regressions are detailed in the "Metabolic model" section of the main notebook, called `code_article.ipynb`.

The data used for the children's part of the human metabolism is detailed in the following folder (details in the notebook): https://archive.softwareheritage.org/browse/origin/directory/?origin_url=https://git.sr.ht/~tfardet/OrgMatt&path=extra/extracted_data&release=v1.0.0

D.3. Data to couple the human and urban metabolisms

Data about urination and defecations times during the day was used to assign a location for the nitrogen and phosphorus excreted on days were a commute occurs. This data, taken from Burgio et al. (1991) and Heaton et al. (1992). All code and data are detailed in the "Individual habits: linking human and urban metabolisms" section of the main notebook, called `code_article.ipynb`.

D.4. Population and sanitation data

Population data are given in their respective folders called `toulouse/` and `amsterdam/` respectively. In the case of the Toulousian area, the raw sanitation data is also provided in `toulouse/listeObj.csv`. For Amsterdam, data from the related articles is provided directly in the main text and in the notebook.